

Proposal: Enhancing Accessibility to Electric Vehicle Charging Stations in Seattle, Washington

By Max Davis & Harsh Malik
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Executive Summary:

As the city of Seattle works towards its decarbonization goals, equal and adequate access to electric vehicle (EV) charging stations is essential to incentivize EV purchases and close the income gap in EV ownership. This step of improving EV infrastructure is crucial for our fight against climate change and addressing racial and economic injustices. Although Seattle is doing a good job in this goal, many Seattlites lack access to public charging stations, particularly in lower-income areas with less access to private chargers.

This proposal examines the coverage of existing EV charging stations in Seattle and identifies areas where more charging stations would be most beneficial. According to our analysis of traffic volume, average annual income, and charging station distribution data, South Seattle has the greatest need for new charging stations. Specifically, the neighborhoods of Roxhill, South Delridge, South Park, Mid-Beacon Hill, and Southeast Rainier Beach are the most underserved. We found the Downtown area and wealthy residential areas to have the greatest coverage. Planning for new public charging stations should prioritize low-income (less than \$42,000 annually) neighborhoods because the higher percentage of apartment buildings and increased burden of costs reduces capacity for private chargers. Additionally, low-income areas with high traffic volume (more than 24,000 vehicles per day on average) should be further prioritized due to their location along major daily commuting routes and subsequent ease of access to the greatest number of people. Investing in these communities will ensure that everyone can make the transition to electric vehicles, no matter where they reside or how much money they have. We have produced a series of maps that illustrate the precise locations of coverage gaps and offer informed recommendations for priority areas for new installations.

By installing more strategically placed public EV charging stations, we will lower the obstacles to going green and encourage more individuals to switch to electric vehicles. Our goal is to assist you in making well-informed decisions that will benefit the people of Seattle, especially those living in underprivileged areas. Thanks to your leadership in giving these areas top priority, Seattle will set the standard for developing a more accessible and sustainable transportation system for everybody. In your ongoing efforts to shape our city's future and create a greener Seattle, we sincerely hope you will consider our suggestions.

Introduction:

The expansion of electric vehicle (EV) charging infrastructure is essential to making Seattle greener. This will also help Seattle become more accessible as the city strives to create a more sustainable future. Many Seattle residents still have trouble getting the necessary charging infrastructure. People may find it challenging to adopt EVs completely, which are essential for lowering carbon emissions and enhancing air quality. This is all caused by disparity in charging accessibility. This issue is important because ensuring that all areas have fair access to EV charging stations is essential to supporting the shift to electric vehicles. If people don't have access to charging stations, then people who wish to transition to electric vehicles will feel like they have few alternatives. This is evidence for people living in low-income communities who might not have the budget to install chargers at home. We believe that improving charging accessibility will not only contribute to cleaner air but also help families transition to EVs more smoothly.

Finding the areas where more charging stations would have the biggest impact is the main goal of our plan. We want to examine areas with heavy traffic and low-income populations that have the least access to infrastructure for charging. By concentrating on these areas, we can guarantee that everyone has access to EV charging stations, ensuring a fair opportunity to make the switch to an electric vehicle. At last, this helps reduce emissions and ensure a cleaner future for all of us. . . .

Methods:

To examine the accessibility of EV charging stations across Seattle, we obtained several datasets from the U.S. Department of Energy and Seattle GeoData:

1. EV Charging Station Locations:

- U.S. Department of Energy Alternative Fuels Data Center (2025).
- This data provides the locations of existing EV charging stations in Seattle.

2. Income Data:

- Seattle GeoData – Per Capita Income and Aggregate Income in the past 12 months (inflation-adjusted dollars) - American Community Survey 2023 (updated 2024).
- This dataset provides detailed income information at the neighborhood level. This allowed us to identify low-income areas that may lack access to charging infrastructure.

3. Network Analysis Data:

- Seattle GeoData – Seattle Street Network Database 2022 (updated 2025).
- The development of network models to examine the accessibility of charging stations inside the city's roadway network is aided by this data.

4. Traffic Data:

- Seattle GeoData – 2022 Traffic Flow Counts – by SDOT (updated 2024).
- This dataset provides traffic volume data that helps identify high-traffic areas.

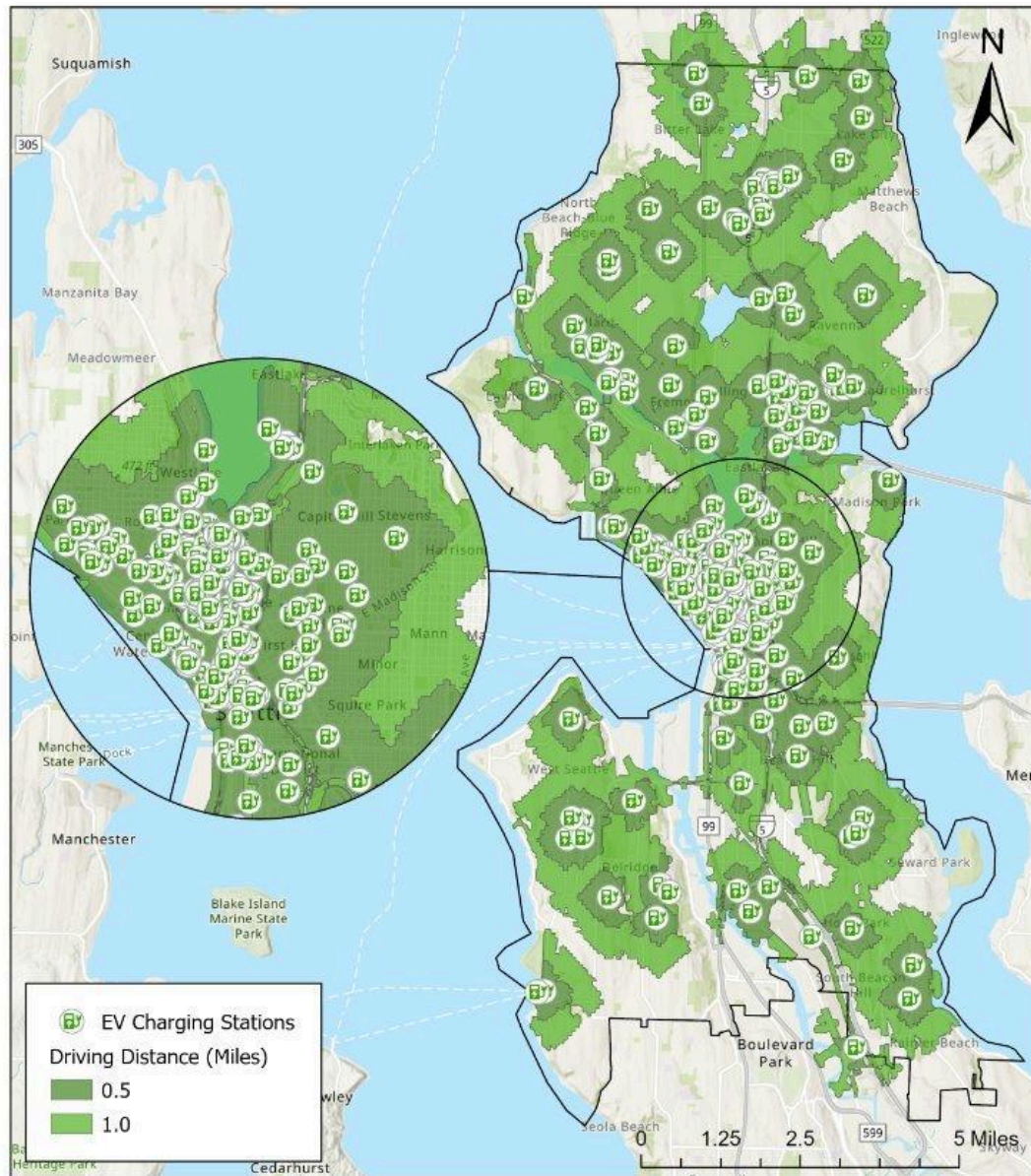
5. Additional Data:

- Seattle GeoData – 2020 Census Tracts (updated 2024).
- Seattle GeoData – Seattle Area Polygon 2022 (updated 2024).
- Seattle GeoData – Neighborhood Map Atlas: Neighborhoods 2020 (updated 2025).
- This source provides demographic and geographical boundaries for analysis.

To locate areas with the greatest need for new public charging stations and highlight disparities in access, we used ArcGIS Pro to conduct a spatial analysis that combined traffic flow and average annual income. Using the results of our analysis, we also provide recommendations for the most advantageous locations for new charging stations to benefit the highest number of citizens possible, especially those who face the most significant barriers to EV adoption.

Driving Distance to EV Charging Stations in Seattle, WA

By Max Davis & Harsh Malik - Intermediate GIS (BIS 344) - UW Bothell - 3/5/2025



Esri, NASA, NGA, USGS, WSU Facilities Services GIS, City of Seattle, King County, WA State Parks GIS, Esri, TomTom, Garmin, SafeGraph, GeoTechnologies, Inc, METI/NASA, USGS, Bureau of Land Management, EPA, NPS, USDA, USFWS

Map 1:

Description: Using network analysis, this map displays the locations in Seattle that are between 0.5 and 1.0 miles away by car from public EV charging stations. The map shows where access is restricted and offers information on how well public charging infrastructure serves certain communities.

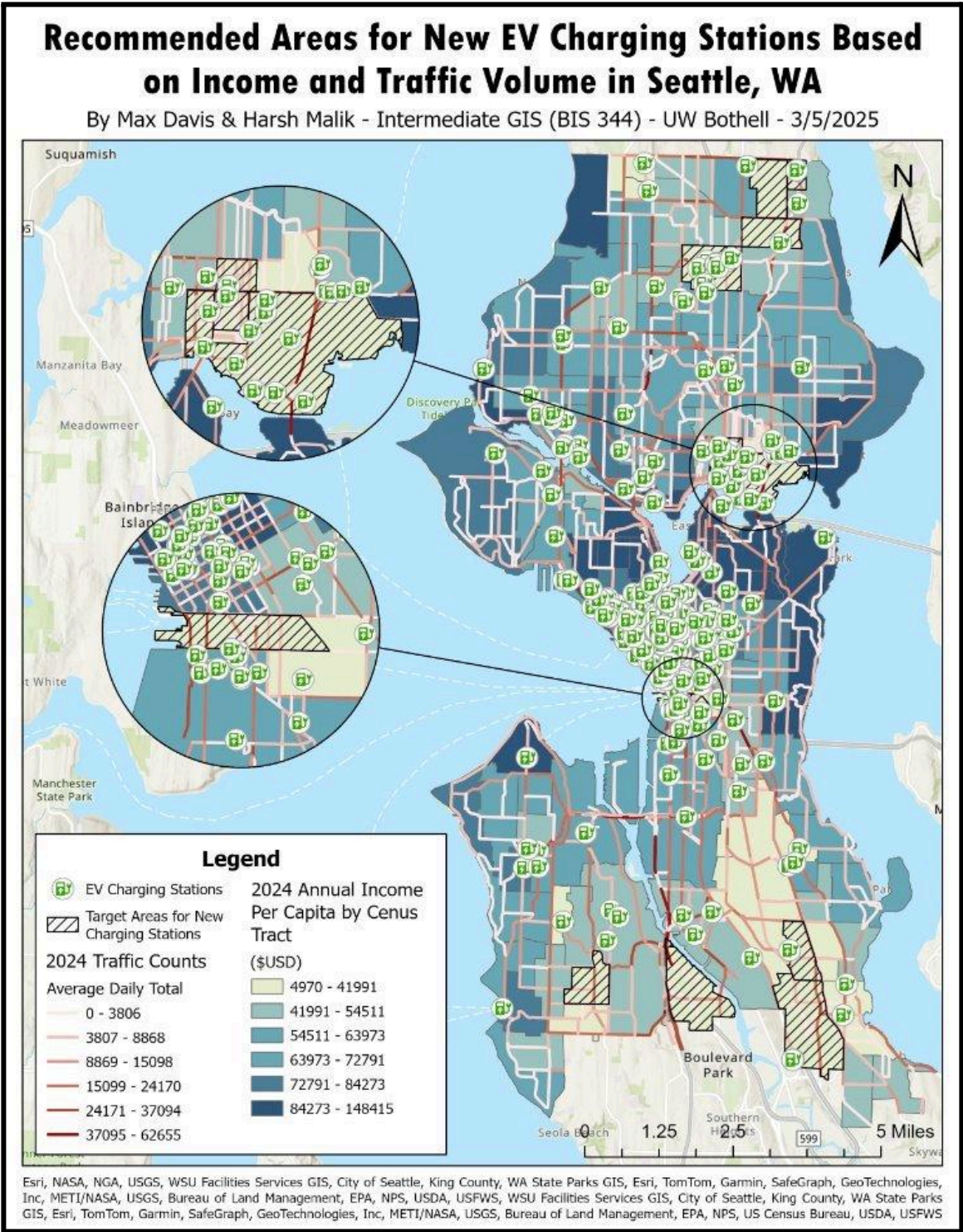
Key Findings:

- Downtown Seattle has the greatest coverage of EV charging stations. Numerous charging points are located within easy driving distance for residents and commuters.
- South Seattle, particularly neighborhoods like Rainier Valley and Delridge, has the least coverage, which highlights a critical gap in accessibility for residents in these areas.

Additional Notes:

- Many unhighlighted regions are high-income residential areas where most people can install a personal charger on their property.

Map 2:



Description: The suggested locations for new EV charging stations in Seattle are indicated on this map according to two important variables: traffic volume and income. Census tracts with low income (household income < \$42,000 annually) and significant traffic volume (ADT > 24,000) are recommended places. The fact that these locations serve both densely populated areas and groups that might lack the means to set up personal chargers at home makes them more crucial.

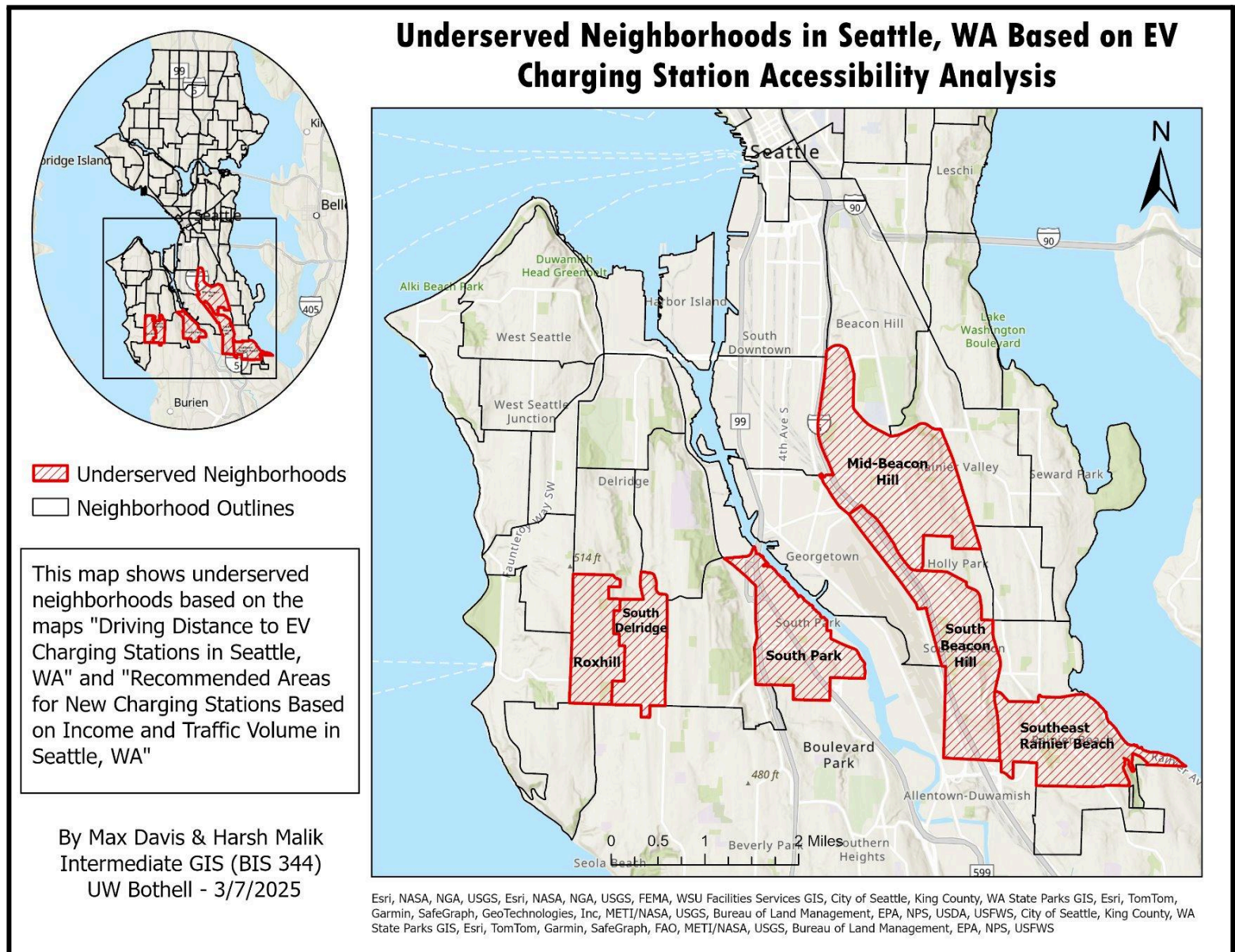
Key Findings:

- Low-income communities are less likely to have access to personal EV chargers, which makes public charging infrastructure critical in these neighborhoods.
- High traffic volume areas are ideal locations for new chargers as they ensure the stations are in convenient spots for a larger number of people.
- The combination of high traffic and low income makes these areas prime candidates for public EV charging stations. This ensures equitable access for those who rely on public infrastructure for EV charging.

Additional Notes:

- Many of the recommended areas are in neighborhoods like South Seattle and Rainier Valley. These areas have significant populations of low-income residents and limited access to private charging options.

Map 3:



Description: The Seattle neighborhoods with the fewest public EV charging facilities are highlighted on this map. Our research indicates that these regions have the greatest need for additional public EV charging stations because of heavy traffic, low income, and a lack of existing charging infrastructure. New installations should be prioritized in the neighborhoods indicated to guarantee fair access to EV charging stations.

Most Underserved Neighborhoods:

- Roxhill
- South Delridge
- South Park
- Mid-Beacon Hill
- Southeast Rainier Beach

Key Observations:

- South Seattle overall has the greatest need for new public EV charging stations. The highlighted neighborhoods have fewer charging stations, low income, and high traffic volume. This makes them prime candidates for infrastructure investment.
- Many of these neighborhoods are home to populations that may not have access to personal EV chargers at home, which makes the need for public chargers even more in demand.

Additional Notes:

- Prioritizing these areas would help ensure that underserved communities have access to the infrastructure they need for a cleaner and more sustainable future

Results and Discussion:

Our analysis showed significant issues in EV charging access across Seattle:

1. **Disproportionate Charging Station Distribution:** Public EV chargers are widely available in downtown Seattle and wealthy areas like Queen Anne and Capitol Hill. Although South Seattle, including areas like Rainier Valley and Delridge, experiences high traffic and has a greater need for public charging stations, it remains underserved.
2. **Underserved Areas:** With a high percentage of low-income residents, South Seattle and the Rainier Valley are the most underserved areas. These locations are perfect for new charging stations because they face major traffic.
3. **Data-Driven Prioritization:** New charger installations should be given priority in low-income regions with median family incomes under \$42,000 annually. We should also give priority to high-traffic areas, especially those with an Average Daily Traffic (ADT) of over 24,000. This would provide convenient access to a larger number of people
4. **Equity considerations:** According to demographic data, there are frequently fewer charging stations in low-income regions. Equal access to clean transportation and environmental justice would be advanced by giving priority to these communities when new installations are being made.

Further Recommendations:

We advise including community input in the design process to make sure we completely comprehend the difficulties users are having with the existing EV charging infrastructure. Information on user experiences and possible weaknesses in the current system can be obtained by compiling insights from community-based resources such as Yelp, Google Reviews, and others. The effectiveness of the current charging stations will also be evaluated with the use of online or in-person community surveys. This will also reveal any issues consumers may be facing that we may not be aware of. This will help guarantee that the new infrastructure is genuinely beneficial and satisfies the needs of all locals. The public will be consulted regarding their suggestions for the layout and positioning of new stations. Through direct community feedback, policymakers can better understand the issues people are facing and make decisions that will improve access and satisfaction for all users. This will ensure a successful transition to electric vehicles. This feedback is essential in identifying issues that may not be immediately apparent.

We would also like to highlight the importance of working toward bipartisan legislation and policies. . . .

Conclusion:

The proposal suggests a focused strategy for expanding Seattle's EV charging station inventory. We can guarantee that all Seattle residents have access to the infrastructure they need for electric vehicle charging by giving priority to low-income and heavily traveled neighborhoods. This is evident in the Rainier Valley and South Seattle. Our plan is based on a detailed study that looks at where more chargers are needed the most by combining information about people, money, and traffic in different areas. Adding more charging stations in neighborhoods that need them the most will help more people use electric cars and help the environment. This will also bring Seattle closer to its goal of being cleaner and greener. This plan is an important first step to make sure everyone in Seattle can switch to electric cars fairly and easily. . . .